

## Advanced (Habanero)

**Q1.** Convert  $x^2 + 6x + 8$  to vertex form.

- (A)  $(x + 3)^2 - 1$
- (B)  $(x + 3)^2 + 1$
- (C)  $(x + 4)^2 - 8$
- (D)  $(x + 2)^2 - 4$
- (E)  $(x + 1)^2 - 7$

**Q2.** Complete the square for  $x^2 - 4x$ .

- (A)  $x^2 - 4x + 4$
- (B)  $x^2 + 4x + 4$
- (C)  $(x - 2)^2$
- (D)  $(x + 2)^2$
- (E)  $x^2 - 4x - 4$

**Q3.** The vertex form of  $x^2 + 2x + 1$  is

- (A)  $(x + 1)^2$
- (B)  $(x + 2)^2 - 1$
- (C)  $(x - 1)^2$
- (D)  $(x - 2)^2 + 1$
- (E)  $x^2 - 2x + 1$

**Q4.** The equation  $y = x^2 + 4x + 4$  can be written in the form  $y = (x + )^2$ .

- (A) 1
- (B) 2
- (C) 3
- (D) 4
- (E) 5

**Q5.** Convert  $y = x^2 + 2x - 6$  to vertex form.

- (A)  $y = (x + 1)^2 - 7$
- (B)  $y = (x - 1)^2 + 7$
- (C)  $y = (x + 2)^2 - 10$
- (D)  $y = (x - 2)^2 + 2$
- (E)  $y = x^2 + 2x + 6$

**Q6.** The vertex of the parabola  $x^2 - 4x - 3$  is

- (A)  $(2, -7)$
- (B)  $(-2, -3)$
- (C)  $(2, 1)$
- (D)  $(-2, -1)$
- (E)  $(-1, -4)$

**Q7.** The equation  $y = x^2 - 6x + 8$  in vertex form is

- (A)  $y = (x - 3)^2 - 1$
- (B)  $y = (x + 3)^2 + 1$
- (C)  $y = (x - 4)^2 + 8$
- (D)  $y = (x + 2)^2 - 4$
- (E)  $y = x^2 - 6x + 9$

**Q8.** The equation  $x^2 - 2x - 1$  in vertex form is  $y =$ 

- (A)  $(x - 1)^2 - 2$
- (B)  $(x + 1)^2 + 2$
- (C)  $(x + 2)^2 - 5$
- (D)  $(x - 2)^2 + 3$
- (E)  $x^2 + 2x + 1$

**Open Ended Questions (FRQ)****Q9.** Convert  $y = x^2 + 8x + 12$  to vertex form and identify the vertex of the parabola.**Q10.** Complete the square to rewrite  $x^2 + 12x + 5$  in vertex form.**Chef's Tips**

1

- Emphasize the importance of completing the square to convert standard form to vertex form.
- Have students interpret the transformed equations to identify the vertex of the parabola.
- Remind students to check their work by plugging in values to ensure accuracy.